Sunil Kumar

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EDUCATION

Duration	Degree/Certificate	College/School	CGPA (10)/%
July 2024 - Present	Ph.D. Civil Engineering (Hydraulics)	IIT Gandhinagar	10.0
July 2014 - August 2018	B.Tech (Civil Engineering)	IIT Patna	7.57
May 2011 - June 2013	Senior Secondary Examination	RBSE	88.80
May 2009 - May 2011	Secondary Examination	RBSE	88.50

PUBLICATIONS

[1] I. M. Tripathi, **S. Kumar**, A. Modi, A. M. Bhat, C. Bhagat, P. K. Mohapatra, and M. L. Kansal, "Towards water sensitivity: A critical review of urban water management strategies," *Water Resources Management*, vol. 40, p. 94, 2026. DOI: 10.1007/s11269-025-04377-2.

DOCTORAL THESIS

Numerical Simulation of Vortex Drop Shaft Hydrodynamics under Multiphase Flow Conditions
(Supervisor: Prof. Pranab K. Mohapatra) *Proposal Defended: Oct 8, 2025*

- **Research Scope:** Numerical and experimental investigation of vortex drop shafts using tangential inlet designs.
- **Methodology:** Utilizing Computational Fluid Dynamics (CFD) to model complex multiphase flow conditions.
- **Experimental Validation:** Designing a physical experimental setup in the hydraulics laboratory using tangential inlets to validate numerical models.
- **Coursework:** Applied Hydraulic Transients, Advanced Hydraulic Engineering, Advanced Engineering Hydrology, Microwave Remote Sensing.

EXPERIENCE

- **Junior Research Fellow (JRF), cNarmada Project, IIT Gandhinagar** *July 2024 – Present*
  - Developing an integrated river basin management plan for the Narmada River.
- **Graduate Teaching Fellow (GTF) IIT Gandhinagar, Gujarat, India** *August 2025 – November 2025*
  - Conducted tutorials and guided 39 M.Tech students in a PG writing course.
- **Teaching Assistant (TA) IIT Gandhinagar, Gujarat, India** *January 2025 – April 2025*
  - Assisted with the undergraduate Fluid Mechanics course for 2nd and 3rd year B.Tech students.
- **Junior Research Fellow (JRF) Water4Change Project, IIT Gandhinagar** *Sept 2022 - July 2024*
  - Contributed to an Indo-Dutch initiative for urban water challenges in India.
- **Summer Internship NWR Jaipur, India** *May 2017 – July 2017*
  - Conducted a study on managing performance in shed installation and initial steps of railway track construction.

RESEARCH & PROJECTS

- **Numerical Simulation of Vortex Drop Shaft Hydrodynamics, IIT Gandhinagar** *Ongoing*
  - Investigating the complex fluid dynamics of vortex drop shafts using experimental methods and advanced numerical modeling (CFD).
  - Modeling multiphase flow in vortex drop shafts to analyze flow behavior and efficiency.
- **Chichlik Pumped Storage Project CFD Analysis (Consultancy Project)** *Oct 2025 - May 2026*
  - Performed end-to-end Computational Fluid Dynamics (CFD) analysis using FLOW-3D HYDRO to assess reservoir intake operating conditions.

- Evaluated velocity fields, pressure distribution, free-surface behavior, and anti-vortex beam performance across multiple generation and pumping scenarios.
- Utilized Civil 3D for CAD geometry preparation and ParaView for advanced post-processing and technical visualization.

- **Assessment of GLOF Susceptibility Across Major Himalayan River Basins** *Jan 2026 - April 2026*
  - Assessed 19,284 glacial lakes across the Indus, Ganga, and Brahmaputra basins using multi-source remote sensing data (Sentinel-1 SAR, ALOS 3D DEM, CHIRPS, and MODIS).
  - Developed basin-wise machine learning models (XGBoost framework) to map susceptibility based on lake dynamics, topography, and climate, achieving high validation R^2 scores (> 0.82).
- **Ground Response Analysis of Soil Deposit, IIT Patna** *July 2017 - May 2018*
  - Analyzed the ground response of soil deposits at the IIT Patna campus under the guidance of Dr. Amarnath Hegde.
  - Observed how response parameters like Peak Ground Acceleration (PGA) and maximum shear strain are affected by variations in input motion and shear wave velocity.
- **Steel and Reinforced Concrete Design Projects, IIT Patna** *Sept 2016 - April 2017*
  - Designed an industrial steel shed for a manufacturing unit using ETABS under Dr. Vaibhav Singhal.
  - Executed the design of a reinforced concrete structure for a three-story framed residential building using SAP2000.
- **Sociology of Development Project, IIT Patna** *Jan 2017 - April 2017*
  - Conducted a water quality check in the nearby area of the IIT Patna campus to assess environmental conditions.

TECHNICAL SKILLS

- **Software & Simulation:** FLOW-3D, FLOW-3D HYDRO, HEC-RAS, ParaView (3D Post-processing), FLOW-3D POST, ANSYS Fluent, DeepSoil, SAP2000, ETABS.
- **Design & GIS:** AutoCAD Civil 3D, AutoCAD 2D Drafting, QGIS, ArcGIS, QSWAT, Google Earth Engine (GEE).
- **Programming:** Python, MATLAB, Java, C/C++, LaTeX.
- **Laboratory & Surveying:** Vectrino+ (ADV), Dumpy Level, Auto Level, Theodolite, Total Station.

ACHIEVEMENTS & AWARDS

- **Inspire Scholarship:** Awarded by the Department of Science & Technology, Government of India.
- **MCM Scholarship:** Awarded Merit-cum-Means Scholarship two times during undergraduate studies.
- **Competition Rank:** Secured 1st rank in Hydraulic Arm designing competition.
- **State Award:** Awarded with a Laptop by the Rajasthan Government for academic excellence.
- **GATE Qualification:** Qualified for the GATE exam in Civil Engineering for eight consecutive years (2018–2025).

REFERENCES

Dr. Pranab K. Mohapatra
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